

Centered Dougall Resonance

Exact Stieltjes Jets, Harmonic Identities,
Polylogarithmic Subtraction, and Zeta Antiderivatives

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Abstract

The incoming Gauss–Hurwitz article asks for a Dougall-type summation with an explicit residue polynomial and complete resonant spectral jets. We supply such a construction for the Rogers–Dougall ${}_5F_4(1)$ sum. Centering the summation variable at half the very-well-poised parameter removes every odd inverse-power correction. The spectral exponent is doubled, so integral and half-integral parameter values lead to two different zeta towers.

At every nonpositive integer parameter, an ordinary, explicitly subtracted series collapses to one short digamma expression. Twice the Gamma-product residue is

$$\frac{(-1)^N}{N!} (d_0)_N (1-b)_N (1-c)_N.$$

The decisive ingredient is a differential identity for the Bernoulli coefficient polynomials. It cancels all lower Hurwitz counterterms, rather than merely leaving a finite but lengthy expression. Every higher spectral jet has an explicit finite formula in Gamma/polygamma data, generalized Stieltjes constants, and Hurwitz-zeta derivatives at negative odd integers. The interlaced half-integer values instead involve nonpositive even integers and no Stieltjes pole.

We derive all-order harmonic identities, including a first-resonance identity containing $\zeta'(-1)$, and a continuous-shift harmonic formula. Every Euler primitive of the first half-parameter harmonic jet has a closed zeta endpoint. A separate finite calculus gives antiderivatives of all collapsed resonance values in the centering parameter. The proofs are accompanied by 485 exact assertions and 293 numerical diagnostics at 70 working decimal digits. Classical summation, parameter-differentiation, and negative-polygamma antecedents are credited explicitly; no global priority or proof-assistant formalization is claimed.

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1 Research target, provenance, and contribution

The canonical ProveIt manuscript connects polylogarithms, harmonic sums, Gamma functions, and Hurwitz-zeta jets. Its incoming article *Gauss–Hurwitz Pole Cancellation* makes a concrete proposal in its Section 9, question 1: pass from Gauss summation to a fixed Dougall-type family whose resonance residues and complete jets can be made explicit [9]. This article answers that specific proposal. It does not classify all generalized hypergeometric summations admitting such a calculus.

The principal result is not a new version of Dougall’s classical summation formula. It is the explicit centered subtraction and cancellation theorem below, together with its derivative and antiderivative consequences. There are three structural differences from the Gauss construction. The natural large- n coordinate is $n + \kappa$; its inverse-power expansion has only even steps; and the exponent is $-1 - 2u$, not $-1 - u$. Consequently the Stieltjes pole has residue $1/2$ as a function of u , while negative half-integer values of u are regular points of the Gamma product.

The central proof is short once its coefficient differential law is found. The surrounding analytic work is nevertheless necessary: the subtracted sum must converge normally, all parameter derivatives must be legitimate, and reciprocal-Gamma zeros must be preserved before differentiation. The article treats each of these points directly. Its conclusions are ordinary analytic identities, not deductions from small numerical residuals.

1.1 Classical inputs and the priority boundary

The Rogers–Dougall very-well-poised ${}_5F_4(1)$ identity is classical and is recorded as DLMF 16.4.9 [1]. Bernoulli expansions for Gamma ratios, Hurwitz-zeta continuation, and the Gamma-zeta derivative relation are also classical [2, 3]. Bühring’s work treats the asymptotic behavior of generalized hypergeometric partial sums, including the constant-term setting relevant to resonance [5]. Thus a finite-part viewpoint on divergent hypergeometric sums is not being introduced here.

Parameter differentiation of hypergeometric identities is an established source of harmonic-number identities. In particular, Chu and Fu treat Dougall–Dixon formulas and harmonic identities [6]; Nicholson gives another account of hypergeometric transformations with harmonic numbers [7]. The polynomial-weighted primitive construction used later belongs to the Hurwitz-zeta and negative-polygamma framework of Espinosa and Moll [8], already present in the canonical integration chapter [11].

The contribution supplied for integration is therefore specific: a parity-adapted coefficient system, its differential cancellation law, the all-integer collapsed formula, a degeneracy-safe all-jet identity, the interlaced even-argument tower, and their explicitly proved specializations. The limited literature inspection does not establish that each individual harmonic specialization is absent from earlier work.

1.2 Source scope and unresolved questions

The manuscript README and integration chapter were inspected, together with the incoming-directory listing and the correspondingly named Gauss–Hurwitz PDF in the Library. The latter’s question on printed page 19 is the target addressed here. The last observed repository head was 16c7e342d7a4a15f59911f322eb90b7f96c2a8b5. The source reads were observations of a changing branch, not an exhaustive audit of that entire commit. Individual file and archive identifiers are recorded in `notes/SOURCE_AUDIT.md`.

The canonical README states that S_4 has an exact proof while S_6 and the revised S_8 candidate remain conjectural [10]. Nothing in this article supplies their missing colored-polylogarithm certificates. No numerical independence claim, new repository commit, or pull request is made. The new results answer one expressly proposed research direction; they do not change unrelated conjecture statuses.

2 The normalized Rogers–Dougall family

2.1 Parameters, powers, and spectral derivatives

Throughout the main construction, fix real parameters

$$\kappa > 0, \quad b > 0, \quad c > 0, \quad d_0 = 1 + 2\kappa - b - c > 0, \quad x_n = n + \kappa. \quad (2.1)$$

The variable u is complex. Powers of x_n use its real logarithm. A prime or a superscript on $\zeta(s, a)$ differentiates its first, *spectral*, variable unless a parameter derivative is displayed. We use

$$\zeta(1 + v, a) = \frac{1}{v} + \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \gamma_m(a) v^m, \quad \gamma_0(a) = -\psi(a). \quad (2.2)$$

The ordinary Stieltjes constants are $\gamma_m = \gamma_m(1)$, with $\gamma_0 = \gamma$ the Euler constant. The notation $(a)_N$ denotes the rising factorial, including $(a)_0 = 1$.

Define the summand by

$$W_n(u) = x_n \frac{\Gamma(n + 2\kappa)\Gamma(n + b)}{\Gamma(n + 1)\Gamma(n + 1 + 2\kappa - b)} \frac{\Gamma(n + c)\Gamma(n + d_0 - u)}{\Gamma(n + 1 + 2\kappa - c)\Gamma(n + b + c + u)}. \quad (2.3)$$

Every reciprocal Gamma factor is interpreted as the entire function $1/\Gamma$. In particular, a summand is allowed to be zero. On $\operatorname{Re} u < d_0$ all numerator Gamma arguments in (2.3) have positive real part, so there are no numerator poles. The fixed denominator arguments are positive by (2.1).

2.2 The classical summation in the required normalization

Put

$$P(u) = \frac{\Gamma(b)\Gamma(c)\Gamma(d_0 - u)}{\Gamma(d_0)\Gamma(b + u)\Gamma(c + u)}, \quad G(u) = \frac{1}{2}\Gamma(u)P(u). \quad (2.4)$$

Then the Rogers–Dougall formula gives

$$\sum_{n=0}^{\infty} W_n(u) = G(u), \quad 0 < \operatorname{Re} u < d_0. \quad (2.5)$$

For clarity, its standard form is

$$\begin{aligned} {}_5F_4 \left(\begin{matrix} a, 1 + a/2, b, c, d \\ a/2, 1 + a - b, 1 + a - c, 1 + a - d \end{matrix} ; 1 \right) \\ = \frac{\Gamma(1 + a - b)\Gamma(1 + a - c)\Gamma(1 + a - d)\Gamma(1 + a - b - c - d)}{\Gamma(1 + a)\Gamma(1 + a - b - c)\Gamma(1 + a - b - d)\Gamma(1 + a - c - d)}. \end{aligned} \quad (2.6)$$

Here $\operatorname{Re}(1 + a - b - c - d) > 0$ [1]. Substitute $a = 2\kappa$ and $d = d_0 - u$. The ratio $(1 + a/2)_n / (a/2)_n = (n + \kappa)/\kappa$ produces the factor x_n . The normalization in (2.3), followed by $\Gamma(1 + 2\kappa) = 2\kappa\Gamma(2\kappa)$, leaves exactly the factor $1/2$ in (2.4).

The sum of the lower hypergeometric parameters minus the sum of the upper ones is $2u$. Thus the usual hypergeometric excess is $2u$, even though the Gamma factor that detects the integral resonances is $\Gamma(u)$. This distinction controls every factor of two below.

3 Centered parity and normally convergent subtraction

3.1 The coefficient polynomials

Set

$$r_1 = \kappa, \quad r_2 = b - \kappa, \quad r_3 = c - \kappa, \quad r_4 = 1 + \kappa - b - c - u. \quad (3.1)$$

In the coordinate $x = x_n$ the summand becomes

$$W_n(u) = x \prod_{i=1}^4 \frac{\Gamma(x + r_i)}{\Gamma(x + 1 - r_i)}. \quad (3.2)$$

Define polynomials ℓ_j and A_j by

$$\ell_j(u) = -\frac{1}{j(2j+1)} \sum_{i=1}^4 B_{2j+1}(r_i), \quad j \geq 1, \quad (3.3)$$

$$\sum_{j \geq 0} A_j(u) y^j = \exp \left(\sum_{j \geq 1} \ell_j(u) y^j \right). \quad (3.4)$$

The second identity is formal in y ; convergence of the infinite asymptotic expansion is not asserted. Its finite recursion is

$$A_0 = 1, \quad A_j = \frac{1}{j} \sum_{r=1}^j r \ell_r A_{j-r}. \quad (3.5)$$

For example,

$$A_1 = \ell_1, \quad A_2 = \ell_2 + \frac{\ell_1^2}{2}, \quad A_3 = \ell_3 + \ell_1 \ell_2 + \frac{\ell_1^3}{6}. \quad (3.6)$$

These expressions are polynomials with rational coefficients in κ, b, c, u .

Lemma 3.1 (Even inverse-power expansion). *For every fixed positive integer K ,*

$$W_n(u) = x_n^{-1-2u} \left(\sum_{j=0}^{K-1} A_j(u) x_n^{-2j} + O(x_n^{-2K}) \right). \quad (3.7)$$

The remainder is locally uniform in the admissible parameters and in u with $\operatorname{Re} u < d_0$. After m spectral derivatives, the error is $O(x_n^{-1-2\operatorname{Re} u-2K}(1 + \log^m x_n))$ on such compact sets.

Proof. The classical logarithmic Gamma-ratio expansion is

$$\begin{aligned} \log \frac{\Gamma(x+r)}{\Gamma(x+1-r)} &\sim (2r-1) \log x \\ &+ \sum_{k \geq 1} \frac{(-1)^{k+1} (B_{k+1}(r) - B_{k+1}(1-r))}{k(k+1)x^k}. \end{aligned}$$

It follows from the shifted Stirling expansion, uniformly for r in compact sets [2]. Since $B_m(1-r) = (-1)^m B_m(r)$, the coefficient is zero when k is odd. For $k = 2j$ it equals $-B_{2j+1}(r)/(j(2j+1))$. Furthermore $\sum_i r_i = 1-u$, so the prefactor x in (3.2) gives total exponent $1 + \sum_i (2r_i - 1) = -1 - 2u$. Summation and formal exponentiation give (3.7) and (3.5).

For the derivative statement, first multiply the difference in (3.7) by x_n^{1+2u} . Its analytic remainder is $O(x_n^{-2K})$ uniformly in a slightly larger parameter neighborhood. Cauchy's estimates give the same order for its fixed spectral derivatives. Differentiating the removed factor x_n^{-1-2u} then contributes only powers of $-2\log x_n$. The finitely many initial summands do not affect the argument. $\square$

Definition 3.2 (Specified subtraction). For $K \geq 1$, define

$$\mathcal{R}_K(u) = \sum_{n=0}^{\infty} \left[W_n(u) - \sum_{j=0}^{K-1} A_j(u) x_n^{-1-2u-2j} \right]. \quad (3.8)$$

The bracket is a single summand; its components need not be separately summable.

Theorem 3.3 (Holomorphic subtraction identity). *The series (3.8) converges absolutely and normally on*

$$-K < \operatorname{Re} u < d_0. \quad (3.9)$$

Every fixed spectral derivative can be taken term by term. On that strip, with removable values understood,

$$\mathcal{R}_K(u) = G(u) - \sum_{j=0}^{K-1} A_j(u) \zeta(1+2u+2j, \kappa). \quad (3.10)$$

Proof. On a compact subset of (3.9), Lemma 3.1 majorizes the bracket and each fixed derivative by a constant times $n^{-1-\epsilon}(1+\log^m n)$ for some $\epsilon > 0$. This proves normal convergence and holomorphy. On the nonempty substrip $0 < \operatorname{Re} u < d_0$, all terms can be summed separately, and (2.5) gives (3.10). The identity theorem extends it throughout (3.9). Any poles on its displayed right side must be removable because the left side is holomorphic. $\square$

This theorem constructs an ordinary convergent function, not an unspecified finite part. In particular, at $u = -N$ the minimal generic choice $K = N + 1$ leaves summands of size $O(n^{-3})$, and its m th spectral derivative has summands $O(n^{-3}(1+\log^m n))$. The estimates serve only to justify the exact identities.

4 The residue polynomial and complete constant-term collapse

Theorem 4.1 (Explicit residue and coefficient differential law). *Let $N \geq 0$ and define*

$$\rho_N = \frac{(-1)^N}{N!} (d_0)_N (1-b)_N (1-c)_N. \quad (4.1)$$

Then

$$A_N(-N) = \rho_N. \quad (4.2)$$

With $D = \partial_\kappa + \partial_b + \partial_c$, acting on the polynomials with $d_0 = 1 + 2\kappa - b - c$, one also has

$$(\partial_u - D)A_N(u) = \sum_{r=1}^N \frac{B_{2r}(\kappa)}{r} A_{N-r}(u). \quad (4.3)$$

The sum is empty when $N = 0$.

Proof. The residue of $G(u)$ at $u = -N$ is $\rho_N/2$, since

$$\frac{\Gamma(b)}{\Gamma(b-N)} = (-1)^N (1-b)_N, \quad \frac{\Gamma(d_0+N)}{\Gamma(d_0)} = (d_0)_N.$$

The Gamma quotient identities hold by polynomial continuation at zeros. In (3.10) with $K = N+1$, only the $j = N$ zeta counterterm can have a pole there. Its residue as a function of u is $A_N(-N)/2$. Holomorphy proves (4.2). Since both sides are polynomials in the fixed parameters, the identity is a polynomial identity, not merely a generic numerical equality.

For the differential law observe that $Dd_0 = 0$. The operator $E = \partial_u - D$ acts on the four variables in (3.1) by

$$Er_1 = -1, \quad Er_2 = Er_3 = Er_4 = 0.$$

Using $B'_j = jB_{j-1}$ in (3.3) gives $E\ell_r = B_{2r}(\kappa)/r$. Apply E to the formal exponential (3.4), multiply its two formal series, and compare the coefficient of y^N . This proves (4.3) at every order. $\square$

Theorem 4.2 (Collapsed value at every integral resonance). *For every integer $N \geq 0$, under (2.1),*

$$\mathcal{R}_{N+1}(-N) = \frac{\rho_N}{2} [\psi(N+1) - \psi(d_0+N) - \psi(b) - \psi(c) + 2\psi(\kappa)]. \quad (4.4)$$

This formula remains valid when b or c belongs to $\{1, \dots, N\}$; its right side is then zero, without any singular digamma factor.

Proof. First assume $b-N$ and $c-N$ are not nonpositive integers. Near $u = -N$, write $u = -N + t$ and expand

$$G(-N+t) = \frac{\rho_N}{2t} + \frac{\rho_N}{2} [\psi(N+1) - \psi(d_0+N) - \psi(b-N) - \psi(c-N)] + O(t). \quad (4.5)$$

The polar counterterm has constant coefficient

$$A_N(-N+t)\zeta(1+2t, \kappa) = \frac{\rho_N}{2t} + \frac{1}{2}A'_N(-N) - \rho_N\psi(\kappa) + O(t).$$

For the remaining terms, the Bernoulli special values give

$$\sum_{j=0}^{N-1} A_j(-N)\zeta(1-2(N-j), \kappa) = -\frac{1}{2} \sum_{r=1}^N \frac{B_{2r}(\kappa)}{r} A_{N-r}(-N).$$

Substitute these three expressions into (3.10). Equation (4.3) cancels the entire Bernoulli sum against $A'_N(-N) - D\rho_N$. The surviving value is the constant coefficient in (4.5), minus $D\rho_N/2$, plus $\rho_N\psi(\kappa)$.

Since $Dd_0 = 0$, logarithmic differentiation of the two remaining rising factorials gives

$$D\rho_N = \rho_N \sum_{r=1}^N \left(\frac{1}{b-r} + \frac{1}{c-r} \right).$$

Finally, $\psi(b) - \psi(b-N) = \sum_{r=1}^N (b-r)^{-1}$, and similarly for c . This proves (4.4) generically. Both the normally convergent left side and the final right side are continuous, indeed locally analytic, throughout (2.1). The final right side contains digamma functions only at positive arguments. Taking limits therefore proves the asserted degenerate cases as well. $\square$

The zeroth resonance is particularly transparent:

$$\sum_{n=0}^{\infty} \left[W_n(0) - \frac{1}{n + \kappa} \right] = \psi(\kappa) - \frac{1}{2} [\gamma + \psi(d_0) + \psi(b) + \psi(c)]. \quad (4.6)$$

At higher resonances the raw subtracted expression contains several Bernoulli polynomials and derivatives of asymptotic coefficients. Theorem 4.2 says that all of them disappear from its final value. The differential law, rather than a finite check at selected N , explains this cancellation.

5 Every spectral jet, including reciprocal-Gamma zeros

5.1 A holomorphic Gamma factor

For $N \geq 0$ define the analytic germ

$$Q_N(t) = \frac{(-1)^N}{N!} \Gamma(1+t) \prod_{r=1}^N (1-t/r)^{-1} P(-N+t). \quad (5.1)$$

The empty product is one. The elementary Gamma recurrence gives

$$G(-N+t) = \frac{Q_N(t)}{2t}, \quad Q_N(0) = \rho_N. \quad (5.2)$$

The factors $1/\Gamma(b-N+t)$ and $1/\Gamma(c-N+t)$ in P remain entire factors. Formula (5.1) therefore defines all jets even when $\rho_N = 0$.

Put $a_j(t) = A_j(-N+t)$. The subtraction theorem and (2.2) yield the following analytic identity of germs:

$$\begin{aligned} \mathcal{R}_{N+1}(-N+t) &= \frac{Q_N(t) - a_N(t)}{2t} - a_N(t) \sum_{k \geq 0} \frac{(-2)^k}{k!} \gamma_k(\kappa) t^k \\ &\quad - \sum_{j=0}^{N-1} a_j(t) \zeta(1 - 2(N-j) + 2t, \kappa). \end{aligned} \quad (5.3)$$

The first quotient is holomorphic because its numerator vanishes at zero. There is no division by ρ_N .

Theorem 5.1 (Finite closure of all integral-resonance jets). *Write $q_r = [t^r]Q_N(t)$ and $a_{j,r} = [t^r]A_j(-N+t)$. For every integer $m \geq 0$,*

$$\begin{aligned} \frac{\mathcal{R}_{N+1}^{(m)}(-N)}{m!} &= \frac{q_{m+1} - a_{N,m+1}}{2} - \sum_{r=0}^m a_{N,r} \frac{(-2)^{m-r} \gamma_{m-r}(\kappa)}{(m-r)!} \\ &\quad - \sum_{j=0}^{N-1} \sum_{r=0}^m a_{j,r} \frac{2^{m-r} \zeta^{(m-r)}(1 - 2(N-j), \kappa)}{(m-r)!}. \end{aligned} \quad (5.4)$$

Thus the zeta spectral arguments outside the Stieltjes part belong to $\{-1, -3, \dots, 1 - 2N\}$. Stieltjes orders are at most m , and the Gamma germ is required only through Taylor degree $m+1$.

Proof. Take the coefficient of t^m in (5.3). All germs are holomorphic near zero, so ordinary Cauchy products apply. The termwise differentiated convergent series represents the same coefficient by Theorem 3.3. The finite formula remains valid at reciprocal-Gamma zeros because (5.1) is their analytic Taylor expansion, not a logarithmic expansion divided by a zero value. $\square$

5.2 Bell-polynomial formulas away from zeros

We normalize complete exponential Bell polynomials by

$$\exp\left(\sum_{r \geq 1} h_r \frac{t^r}{r!}\right) = \sum_{r \geq 0} Y_r(h_1, \dots, h_r) \frac{t^r}{r!}, \quad Y_0 = 1. \quad (5.5)$$

When $b - N$ and $c - N$ avoid nonpositive integers, the logarithmic coefficients of Q_N are

$$\begin{aligned} h_{N,r} = & \psi^{(r-1)}(1) + (r-1)! H_N^{(r)} + (-1)^r \psi^{(r-1)}(d_0 + N) \\ & - \psi^{(r-1)}(b - N) - \psi^{(r-1)}(c - N), \end{aligned} \quad (5.6)$$

where $H_N^{(r)} = \sum_{j=1}^N j^{-r}$. Consequently

$$q_r = \frac{\rho_N}{r!} Y_r(h_{N,1}, \dots, h_{N,r}). \quad (5.7)$$

This convenient form is not the definition at a zero of ρ_N . The safe computational alternative is to shift every problematic reciprocal Gamma factor to a positive argument:

$$\frac{1}{\Gamma(w+t)} = \frac{\prod_{j=0}^{L-1} (w+j+t)}{\Gamma(w+L+t)}, \quad w+L > 0. \quad (5.8)$$

Expand the polynomial numerator and the nonzero Gamma germ separately. The code implements precisely this operation.

5.3 The zero-excess family in closed Bell form

At $N = 0$, $A_0 = 1$ and $Q_0(0) = 1$. Set

$$h_r = \psi^{(r-1)}(1) + (-1)^r \psi^{(r-1)}(d_0) - \psi^{(r-1)}(b) - \psi^{(r-1)}(c). \quad (5.9)$$

The general formula simplifies to

$$\mathcal{R}_1^{(m)}(0) = \frac{Y_{m+1}(h_1, \dots, h_{m+1})}{2(m+1)} - (-2)^m \gamma_m(\kappa). \quad (5.10)$$

For instance,

$$\begin{aligned} \mathcal{R}_1(0) &= h_1/2 - \gamma_0(\kappa), \\ \mathcal{R}_1'(0) &= (h_1^2 + h_2)/4 + 2\gamma_1(\kappa), \\ \mathcal{R}_1''(0) &= (h_1^3 + 3h_1h_2 + h_3)/6 - 4\gamma_2(\kappa). \end{aligned} \quad (5.11)$$

No integer-relation search is involved in these identities.

6 Half-integral values and exact polynomial degeneracies

6.1 The interlaced even-argument tower

Theorem 6.1 (Half-integral spectral values). *For $N \geq 0$, put $u_* = -N - \frac{1}{2}$. Then*

$$\mathcal{R}_{N+1}(u_*) = G(u_*) + \sum_{j=0}^N A_j(u_*) \frac{B_{2(N-j)+1}(\kappa)}{2(N-j)+1}. \quad (6.1)$$

Every fixed spectral derivative at u_* is a finite expression in Gamma jets and Hurwitz-zeta derivatives at $0, -2, \dots, -2N$; no Stieltjes Laurent coefficient is needed.

Proof. At u_* the counterterm spectral argument is $2(j - N)$, never one. Also $\Gamma(u_*)$ is finite. Substitute directly in (3.10) and use $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$. Ordinary differentiation of the same holomorphic identity gives the jet statement. Here the minimal subtraction has terms $O(n^{-2})$, and the derivatives have the corresponding logarithmic factors. $\square$

The first case is

$$\mathcal{R}_1(-\tfrac{1}{2}) = \kappa - \tfrac{1}{2} - \sqrt{\pi} \frac{\Gamma(b)\Gamma(c)\Gamma(d_0 + \tfrac{1}{2})}{\Gamma(d_0)\Gamma(b - \tfrac{1}{2})\Gamma(c - \tfrac{1}{2})}. \quad (6.2)$$

The different parity of the two zeta towers is a consequence of centered very-well-poised symmetry, not a rule that applies to arbitrary Gamma ratios.

6.2 The symmetric half-parameter line

Take $\kappa = b = c = \tfrac{1}{2}$, so $d_0 = 1$. The summand simplifies to

$$W_n(u) = \frac{\Gamma(n+1-u)}{(n+\tfrac{1}{2})\Gamma(n+1+u)}. \quad (6.3)$$

At negative integer and negative half-integer parameters, the Gamma recurrence gives two exact finite products. Writing $x = n + \tfrac{1}{2}$,

$$W_n(-N) = \frac{1}{x} \prod_{r=1}^N \left(x^2 - (r - \tfrac{1}{2})^2 \right), \quad (6.4)$$

$$W_n(-N - \tfrac{1}{2}) = \prod_{r=1}^N (x^2 - r^2). \quad (6.5)$$

These include any initial zero summands, by reciprocal-Gamma continuation. Their asymptotic expansions terminate identically. In particular,

$$\mathcal{R}_{N+1}(-N) = \mathcal{R}_{N+1}(-N - \tfrac{1}{2}) = 0 \quad \text{on this parameter line.} \quad (6.6)$$

For the integer case,

$$A_j(-N) = (-1)^j e_j \left((\tfrac{1}{2})^2, (\tfrac{3}{2})^2, \dots, (N - \tfrac{1}{2})^2 \right), \quad (6.7)$$

with $A_j(-N) = 0$ for $j > N$. Its final coefficient is $\rho_N = (-1)^N (\tfrac{1}{2})_N^2$.

Proposition 6.2 (An additional exact polynomial locus). *If $N \geq 1$ and $b = m \in \{1, \dots, N\}$, then, under (2.1),*

$$W_n(-N) = x_n \prod_{r=1}^{m-1} (x_n^2 - (\kappa - r)^2) \prod_{r=1}^{N-m} (x_n^2 - (\kappa - c + r)^2). \quad (6.8)$$

Thus its centered expansion ends at $j = N - 1$ and $\mathcal{R}_{N+1}(-N) = 0$ term by term. There is a symmetric assertion with b and c exchanged.

Proof. In (2.3), combine the first pair of relevant Gamma quotients as

$$\frac{\Gamma(n+m)}{\Gamma(n+1)} \frac{\Gamma(n+2\kappa)}{\Gamma(n+1+2\kappa-m)} = \prod_{r=1}^{m-1} (n+r)(n+2\kappa-r).$$

The other pair, with $q = N - m$, is

$$\frac{\Gamma(n+c)}{\Gamma(n+c-q)} \frac{\Gamma(n+1+2\kappa-c+q)}{\Gamma(n+1+2\kappa-c)} = \prod_{r=1}^q (n+c-r)(n+2\kappa-c+r).$$

Center each pair at x_n . Polynomial continuation handles all zero reciprocal Gamma values. This proves (6.8) and the remaining conclusions. $\square$

A specialized polynomial identity cannot be differentiated in u after u has been fixed. The transverse spectral derivatives contain information lost by that specialization. The next section gives a concrete example.

7 Exact harmonic-number identities

7.1 The general all-order harmonic form

For $r \geq 1$, let

$$g_{n,r} = (-1)^r \psi^{(r-1)}(n+d_0) - \psi^{(r-1)}(n+b+c). \quad (7.1)$$

Differentiating the logarithm of (2.3) at zero gives

$$\partial_u^m W_n(u)|_{u=0} = W_n(0) Y_m(g_{n,1}, \dots, g_{n,m}). \quad (7.2)$$

Hence (5.10) is the ordinary convergent identity

$$\begin{aligned} \sum_{n=0}^{\infty} \left[W_n(0) Y_m(g_{n,1}, \dots, g_{n,m}) - \frac{(-2 \log x_n)^m}{x_n} \right] \\ = \frac{Y_{m+1}(h_1, \dots, h_{m+1})}{2(m+1)} - (-2)^m \gamma_m(\kappa). \end{aligned} \quad (7.3)$$

The polygamma differences are generalized harmonic numbers. With $H_n^{(r)}(a) = \sum_{j=0}^{n-1} (j+a)^{-r}$,

$$\psi^{(r-1)}(n+a) = \psi^{(r-1)}(a) + (-1)^{r-1} (r-1)! H_n^{(r)}(a). \quad (7.4)$$

Thus (7.3) is an explicit all-order family of Gamma-weighted harmonic identities. The compact normal convergence proof already justifies every coefficient interchange.

7.2 A half-parameter family without Gamma weights

On the symmetric line of (6.3), $W_n(0) = 1/x_n$. Put $L = \log 2$ and $A = \gamma + 2L$. The right-side Bell data become

$$h_1 = 2A, \quad h_r = \begin{cases} -2(2^r - 2)(r-1)! \zeta(r), & r \geq 2 \text{ even}, \\ 2(2^r - 1)(r-1)! \zeta(r), & r \geq 3 \text{ odd}. \end{cases} \quad (7.5)$$

The left-side data are

$$g_{n,1} = -2(H_n - \gamma), \quad g_{n,r} = \begin{cases} 0, & r \geq 2 \text{ even}, \\ 2(r-1)!(\zeta(r) - H_n^{(r)}), & r \geq 3 \text{ odd}. \end{cases} \quad (7.6)$$

Here $H_n = \sum_{j=1}^n j^{-1}$ and $H_0 = 0$. Substitution in (7.3) gives every order in ordinary harmonic numbers and zeta tails, with no Gamma weights remaining.

At order one this yields

$$\sum_{n=0}^{\infty} \frac{H_n - \gamma - \log(n + \frac{1}{2})}{n + \frac{1}{2}} = \frac{\zeta(2) - \gamma^2 - 2L^2 - 2\gamma_1}{2}. \quad (7.7)$$

At order two it gives

$$\sum_{n=0}^{\infty} \frac{(H_n - \gamma)^2 - \log^2(n + \frac{1}{2})}{n + \frac{1}{2}} = \frac{A^3}{3} - A\zeta(2) + \frac{7}{6}\zeta(3) - \gamma_2(\frac{1}{2}). \quad (7.8)$$

For reference, differentiation of $\zeta(s, \frac{1}{2}) = (2^s - 1)\zeta(s)$ at its Laurent pole gives

$$\begin{aligned} \gamma_1(\frac{1}{2}) &= \gamma_1 - 2\gamma L - L^2, \\ \gamma_2(\frac{1}{2}) &= \gamma_2 - 4L\gamma_1 + 2\gamma L^2 + \frac{2}{3}L^3. \end{aligned} \quad (7.9)$$

The first summands in (7.7) are $O(n^{-3})$; those in (7.8) are $O(n^{-3} \log n)$.

7.3 A first-resonance identity with a revived head

The same half-parameter family has

$$A_1(u) = \frac{u(4u^2 - 1)}{12}, \quad A_1(-1) = -\frac{1}{4}, \quad A_1'(-1) = \frac{11}{12}, \quad A_1''(-1) = -2. \quad (7.10)$$

For $n \geq 1$,

$$\begin{aligned} W_n(-1) &= \frac{n(n+1)}{n + \frac{1}{2}}, \\ W_n'(-1) &= \frac{n(n+1)}{n + \frac{1}{2}} \left[-2(H_n - \gamma) + \frac{1}{n(n+1)} \right]. \end{aligned} \quad (7.11)$$

The initial term must be computed separately. In a local coordinate t ,

$$W_0(-1+t) = \frac{2\Gamma(2-t)}{\Gamma(t)} = 2t + O(t^2). \quad (7.12)$$

Thus $W_0(-1) = 0$ but $W_0'(-1) = 2$. After subtracting the derivative of the two counterterms, the initial bracket is $1/6$.

Corollary 7.1 (An explicit $\zeta'(-1)$ harmonic identity). *One has*

$$\begin{aligned} \frac{1}{6} + \sum_{n=1}^{\infty} \left[\frac{2n(n+1)}{n + \frac{1}{2}} (\log(n + \frac{1}{2}) - H_n + \gamma) + \frac{1}{12(n + \frac{1}{2})} \right] \\ = \zeta'(-1) + \frac{\zeta(2)}{4} - \frac{\gamma^2}{4} - \frac{\gamma_1}{2} - \frac{L^2}{2} + \frac{\gamma}{12} + \frac{L}{4}. \end{aligned} \quad (7.13)$$

Proof. The bracket is $\partial_u \mathcal{R}_2(u)|_{u=-1}$ by (7.10)–(7.12). For this specialization,

$$Q_1(t) = -\frac{\pi\Gamma(1+t)\Gamma(1-t)}{\Gamma(-\frac{1}{2}+t)^2}. \quad (7.14)$$

Insert its Taylor coefficients and (7.10) into (5.4) with $N = m = 1$. Direct simplification gives

$$-\frac{A^2}{4} + \frac{A}{12} + \frac{\zeta(2)}{4} - \frac{\gamma_1(\frac{1}{2})}{2} - 2\zeta'(-1, \frac{1}{2}).$$

Use (7.9) and $\zeta'(-1, \frac{1}{2}) = -\zeta'(-1)/2 - L/24$. This is precisely the right side of (7.13). Normal convergence of the differentiated subtraction proves ordinary convergence of the printed series. $\square$

The values of the left sides of (7.7), (7.8), and (7.13) begin, respectively,

$$0.24824090308572918127, \quad -0.27295567580201109333, \quad 0.18008740839086106049.$$

These decimals are diagnostic illustrations of the proved formulas.

8 Continuous shifts and exact Euler-primitive endpoints

For $a > 0$ and integer $p \geq 1$, let

$$S_p(a) = \sum_{n=0}^{\infty} \frac{H_n - \gamma - \log(n+a)}{(n+a)^p}. \quad (8.1)$$

The numerator is $O(n^{-1})$, locally uniformly in a , so these are ordinary absolutely convergent sums even at $p = 1$.

Theorem 8.1 (A continuous-shift harmonic closure). *For every $a > 0$,*

$$S_1(a) = \frac{\zeta(2, a) - \psi(a)^2}{2} - \gamma_1(a) - \zeta(2). \quad (8.2)$$

For every integer $p \geq 2$,

$$\begin{aligned} S_p(a) &= \psi(a)\zeta(p, a) + \frac{p}{2}\zeta(p+1, a) + \zeta'(p, a) \\ &\quad - \frac{1}{2} \sum_{j=2}^{p-1} \zeta(j, a)\zeta(p+1-j, a). \end{aligned} \quad (8.3)$$

An empty sum is zero.

Proof. Begin with a classical beta differentiation. The expansion at $v = 0$ is

$$B(a, v) = \frac{1}{v} - (\gamma + \psi(a)) + \frac{v}{2}((\gamma + \psi(a))^2 + \zeta(2) - \psi'(a)) + O(v^2). \quad (8.4)$$

For $v > 0$, differentiation under Euler's beta integral gives $\partial_a \partial_v B(a, v)$. At $v = 0$ this mixed derivative remains an ordinary convergent integral: its integrand is $t^{a-1} \log t \log(1-t)/(1-t)$, which is integrable at both endpoints. Using $\sum_{n \geq 0} H_n t^n = -\log(1-t)/(1-t)$, it follows that

$$\sum_{n \geq 0} \frac{H_n}{(n+a)^2} = (\psi(a) + \gamma)\psi'(a) - \frac{1}{2}\psi''(a). \quad (8.5)$$

The integral interchange can also be justified by monotone convergence, since its expanded summands are nonnegative for $a > 0$.

Differentiate (8.5) $p - 2$ times in a , and multiply by $(-1)^{p-2}/(p-1)!$. The series derivatives converge normally on compact subintervals of $a > 0$. Substituting $\psi^{(r)}(a) = (-1)^{r+1}r!\zeta(r+1, a)$ gives

$$\sum_{n \geq 0} \frac{H_n}{(n+a)^p} = (\psi(a) + \gamma)\zeta(p, a) + \frac{p}{2}\zeta(p+1, a) - \frac{1}{2} \sum_{j=2}^{p-1} \zeta(j, a)\zeta(p+1-j, a). \quad (8.6)$$

For the last sum, the Leibniz coefficient before symmetrization is $(p-j)/(p-1)$; the coefficient paired under $j \mapsto p+1-j$ adds to one. Subtract $\gamma\zeta(p, a)$ and use $\sum \log(n+a)/(n+a)^p = -\zeta'(p, a)$ to obtain (8.3).

To prove (8.2), termwise differentiation gives $S'_1(a) = -\zeta(2, a) - S_2(a)$. Differentiating (2.2) in a and using $\partial_a \zeta(s, a) = -s\zeta(s+1, a)$ gives

$$\gamma'_1(a) = \zeta(2, a) + \zeta'(2, a). \quad (8.7)$$

Consequently the derivative of $\frac{1}{2}(\zeta(2, a) - \psi(a)^2) - \gamma_1(a)$ is also $-\zeta(2, a) - S_2(a)$. Their difference is constant on $a > 0$. Evaluate it at $a = 1/2$ using (7.7) and (7.9); the constant is $-\zeta(2)$. $\square$

The beta-differentiation method and the underlying zeta identities are classical. Their role here is to evaluate an entire primitive family of the newly organized spectral identity, not to assert literature priority for (8.6).

At the half shift, (8.3) reduces entirely to Riemann zeta values and derivatives:

$$\begin{aligned} S_p(\tfrac{1}{2}) &= (2^p - 1)\zeta'(p) - ((2^p - 1)\gamma + (2^p - 2)L)\zeta(p) \\ &\quad + \frac{p}{2}(2^{p+1} - 1)\zeta(p+1) \\ &\quad - \frac{1}{2} \sum_{j=2}^{p-1} (2^j - 1)(2^{p+1-j} - 1)\zeta(j)\zeta(p+1-j), \quad p \geq 2. \end{aligned} \quad (8.8)$$

In particular,

$$S_2(\tfrac{1}{2}) = 7\zeta(3) - (3\gamma + 2L)\zeta(2) + 3\zeta'(2), \quad (8.9)$$

$$S_3(\tfrac{1}{2}) = \frac{45}{4}\zeta(4) - (7\gamma + 6L)\zeta(3) + 7\zeta'(3). \quad (8.10)$$

Section 9 explains exactly why these are endpoint values of successive normalized Euler antiderivatives. Notice the change of coordinates: the unintegrated endpoint uses γ_1 , but every positive primitive order in this particular first-jet family uses positive-integer zeta values and their first derivatives instead.

9 Polylogarithmic subtraction and normalized primitives

9.1 Generating functions with their endpoint values retained

For $|z| < 1$ put

$$F(u, z) = \sum_{n \geq 0} W_n(u)z^n, \quad \Phi(z, s, a) = \sum_{n \geq 0} \frac{z^n}{(n+a)^s}. \quad (9.1)$$

The latter is Lerch's transcendent [4]. Define

$$\mathcal{R}_K(u, z) = F(u, z) - \sum_{j=0}^{K-1} A_j(u) \Phi(z, 1 + 2u + 2j, \kappa). \quad (9.2)$$

Equivalently, it is the generating series of the brackets in (3.8).

Theorem 9.1 (Endpoint and spectral-jet lifting). *On compact subsets of $-K < \operatorname{Re} u < d_0$, the bracketed series defining $\mathcal{R}_K(u, z)$ and all its fixed spectral derivatives converge uniformly for $|z| \leq 1$. Consequently*

$$\partial_u^m \mathcal{R}_K(u, 1) = \mathcal{R}_K^{(m)}(u). \quad (9.3)$$

In particular, (4.4) and (5.4) give exact endpoint values of the corresponding subtracted hypergeometric–Lerch identities.

Proof. Use the summable majorants in Theorem 3.3 and $|z^n| \leq 1$. Uniform convergence gives continuity on the closed disk, and the same argument applies to every fixed spectral derivative. This is stronger than merely assigning an Abel value to a divergent series. The unsubtracted F and the individual Lerch terms need not have finite endpoint values in this range. $\square$

At generic parameters the hypergeometric form is

$$F(u, z) = W_0(u) {}_5F_4 \left(\begin{matrix} 2\kappa, 1 + \kappa, b, c, d_0 - u \\ \kappa, 1 + 2\kappa - b, 1 + 2\kappa - c, b + c + u \end{matrix}; z \right). \quad (9.4)$$

At a singular lower parameter this product can look like zero times an undefined hypergeometric expression. The coefficient definition (9.1), with reciprocal-Gamma factors, is authoritative there.

9.2 Rational centering converts the counterterms to polylogarithms

Let $\kappa = p/q$, $1 \leq p \leq q$, and $\omega = \exp(2\pi i/q)$. For $0 < z < 1$, the root-of-unity filter gives

$$\Phi(z, s, p/q) = q^{s-1} z^{-p/q} \sum_{\ell=0}^{q-1} \omega^{-p\ell} \operatorname{Li}_s(\omega^\ell z^{1/q}). \quad (9.5)$$

To prove it, expand each polylogarithm in its defining series. The finite Fourier sum is q when its summation index is congruent to p modulo q , and zero otherwise. The remaining terms are exactly those in (9.1). Analytic continuation in the punctured disk, using a consistent root, gives the same single-valued analytic combination; the apparent singularity at $z = 0$ cancels. Rational $\kappa > 1$ is reduced by the finite shift formula

$$\Phi(z, s, a + M) = z^{-M} \left(\Phi(z, s, a) - \sum_{n=0}^{M-1} \frac{z^n}{(n + a)^s} \right). \quad (9.6)$$

For spectral derivatives, differentiation of the factor q^{s-1} must be retained. If $s = s_0 + 2u$, then

$$\partial_u^m \Phi(z, s, p/q) = 2^m q^{s-1} z^{-p/q} \sum_{\ell=0}^{q-1} \omega^{-p\ell} (\partial_s + \log q)^m \operatorname{Li}_s(\omega^\ell z^{1/q}). \quad (9.7)$$

Thus every rational-center counterterm and every counterterm jet has an explicit finite polylogarithmic representation. This statement concerns the subtraction terms; it does not assert a general reduction of the remaining ${}_5F_4$ function to classical polylogarithms.

9.3 Euler derivatives and genuine normalized antiderivatives

Define

$$\Theta_\kappa = z \frac{d}{dz} + \kappa, \quad J_\kappa f(z) = z^{-\kappa} \int_0^z t^{\kappa-1} f(t) dt. \quad (9.8)$$

For a power series $f(z) = \sum f_n z^n$, this is the analytic operator $J_\kappa f = \sum f_n z^n / (n + \kappa)$. It is the unique analytic inverse of Θ_κ at zero, and $d(z^\kappa J_\kappa f)/dz = z^{\kappa-1} f(z)$ is an ordinary antiderivative identity. For integers $r \geq 1$,

$$J_\kappa^r f(z) = \frac{z^{-\kappa}}{(r-1)!} \int_0^z t^{\kappa-1} (\log(z/t))^{r-1} f(t) dt. \quad (9.9)$$

Initially one may integrate on $0 < z < 1$; the power-series definition fixes the analytic continuation without a branch ambiguity. It follows that

$$\Theta_\kappa \Phi(z, s, \kappa) = \Phi(z, s-1, \kappa), \quad J_\kappa^r \Phi(z, s, \kappa) = \Phi(z, s+r, \kappa). \quad (9.10)$$

Apply these operators coefficientwise to (9.2). Spectral derivatives commute with J_κ^r on compact parameter sets, and endpoint continuity for the bracketed primitives follows from absolute convergence. Euler differentiation inside the disk is unrestricted; any endpoint claim after differentiation additionally requires the resulting coefficient series to be summable.

The first primitive of the generic hypergeometric side cancels the very-well-poised pair:

$$J_\kappa F(u, z) = \frac{W_0(u)}{\kappa} {}_4F_3 \left(\begin{matrix} 2\kappa, b, c, d_0 - u \\ 1 + 2\kappa - b, 1 + 2\kappa - c, b + c + u \end{matrix}; z \right). \quad (9.11)$$

Again the original coefficient series gives its value at exceptional lower parameters.

For the first spectral jet on the symmetric half-parameter line,

$$-\frac{1}{2} \partial_u \mathcal{R}_1(0, z) = \sum_{n \geq 0} \frac{H_n - \gamma - \log(n + \frac{1}{2})}{n + \frac{1}{2}} z^n. \quad (9.12)$$

Hence all its normalized Euler-primitive endpoints are explicitly known:

$$-\frac{1}{2} \left[J_{1/2}^r \partial_u \mathcal{R}_1(0, z) \right]_{z=1} = S_{r+1}(\tfrac{1}{2}), \quad r \geq 0. \quad (9.13)$$

Equations (7.7) and (8.8) evaluate the entire family. This is a closed endpoint theorem for this specific first jet, not an assertion that arbitrary primitive orders of all higher Dougall jets have already been reduced.

10 Antiderivatives of all collapsed resonance values

The centering parameter gives a different antiderivative problem. Here b, c, N are fixed, while κ varies within (2.1). Theorem 4.2 makes this problem finite because ρ_N is a polynomial in κ .

10.1 A normalized Hurwitz-zeta ladder

For integers $j \geq 0$ set

$$\mathcal{P}_j(x) = \frac{\zeta'(-j, x) + H_j \zeta(-j, x)}{j!}, \quad H_0 = 0. \quad (10.1)$$

Then

$$\mathcal{P}'_0(x) = \psi(x), \quad \mathcal{P}'_j(x) = \mathcal{P}_{j-1}(x) \quad (j \geq 1). \quad (10.2)$$

Indeed, Lerch's classical identity $\zeta'(0, x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi)$ proves the first relation. For the second, differentiate in the parameter to obtain

$$\partial_x \zeta'(-j, x) = j \zeta'(1-j, x) - \zeta(1-j, x), \quad \partial_x \zeta(-j, x) = j \zeta(1-j, x),$$

and use $H_j - 1/j = H_{j-1}$. These normalized negative-polygamma coordinates are classical in substance [8, 3].

For a polynomial p , define

$$\mathfrak{I}[p](x) = \sum_{j=0}^{\deg p} (-1)^j p^{(j)}(x) \mathcal{P}_j(x), \quad \mathfrak{I}[0] = 0. \quad (10.3)$$

Adjacent terms telescope under differentiation, giving

$$\frac{d}{dx} \mathfrak{I}[p](x) = p(x) \psi(x). \quad (10.4)$$

This is a finite formula rather than an unevaluated repeated integral.

Theorem 10.1 (A finite primitive at every resonance). *Let*

$$\begin{aligned} C_N &= \frac{(-1)^N}{N!} (1-b)_N (1-c)_N, & p_N(\kappa) &= C_N (1+2\kappa-b-c)_N, \\ y &= 1+2\kappa-b-c+N, & q_N(y) &= (y-N)_N, \\ B_N &= \psi(N+1) - \psi(b) - \psi(c). \end{aligned} \quad (10.5)$$

Choose any polynomial P_N^{poly} with derivative p_N . An antiderivative, in κ , of $\mathcal{R}_{N+1}(-N)$ is

$$\frac{B_N}{2} P_N^{\text{poly}}(\kappa) + \mathfrak{I}[p_N](\kappa) - \frac{C_N}{4} \mathfrak{I}[q_N](y). \quad (10.6)$$

The special-function coordinates required are only $\zeta'(-j, \kappa)$, $\zeta'(-j, y)$ and Bernoulli polynomials, $0 \leq j \leq N$, together with the displayed constant digamma values.

Proof. Since $p_N(\kappa) = \rho_N = C_N q_N(y)$ and $y' = 2$, differentiating (10.6) by (10.4) gives

$$\frac{B_N}{2} \rho_N + \rho_N \psi(\kappa) - \frac{\rho_N}{2} \psi(y).$$

This is exactly (4.4). The Bernoulli assertion follows from $\zeta(-j, x) = -B_{j+1}(x)/(j+1)$. If $C_N = 0$, the displayed expression is constant and the identity remains valid. $\square$

At $N = 0$, the formula reduces, up to a constant, to

$$\log \Gamma(\kappa) - \frac{1}{4} \log \Gamma(d_0) - \frac{\kappa}{2} (\gamma + \psi(b) + \psi(c)). \quad (10.7)$$

At $N = 1$ only $\zeta'(0, \cdot)$ and $\zeta'(-1, \cdot)$ occur; at each subsequent resonance precisely one further level is sufficient. The theorem concerns zeroth spectral order at all resonances. Mixed κ -primitives of arbitrary spectral jets are a separate reduction problem, listed below.

11 Audit, safeguards, and reproducible evidence

11.1 What is, and is not, an erratum

The inspected canonical statements already distinguish analytic proofs, exact finite checks, and numerical residuals. No additional specific mathematical error in those inspected statements was established by this investigation. In particular, the outgoing package does not relabel a classical identity as an error or silently change the status of an unresolved Gaussian candidate.

There are, however, four essential corrections to a *naive transplant* of the Gauss construction. They are integration safeguards, not accusations that the existing manuscript makes these mistakes.

The doubled excess. The counterterm is $\zeta(1 + 2u + 2j, \kappa)$, with residue $1/2$ in u . Replacing it by $\zeta(1 + u + j, \kappa)$ would destroy the pole cancellation and all derivative factors. The classical ${}_5F_4$ excess is $2u$.

The centering and parity. Only the coordinate $x = n + \kappa$ has the complementary pairs used in (3.2). An expansion in n generally has odd inverse powers. Omitting them without making the centering change is invalid.

Zeros are not zero jets. Equation (7.12) gives a concrete counterexample to dropping a summand that vanishes at the expansion point. Its contribution to the first derivative is 2, and its complete subtracted contribution is $1/6$. Likewise, the Bell formula (5.7) must not be used as zero times divergent polygammas. Formula (5.8) is the replacement.

An asymptotic tail is not an interval certificate. The asymptotic coefficients are used in finite number in both the proof and the numerical checks. Extending the asymptotic series to infinitely many terms at fixed n is not justified. Numerical agreement between finite asymptotic-tail evaluations does not constitute a rigorous error enclosure or establish arithmetic independence.

11.2 Exact finite verification

The delivered `code/verify_exact.py` completed **485 exact assertions**. They include Bernoulli reflection through degree 24; generic logarithmic differential identities through $j = 6$; generic coefficient and residue identities through $N = 3$; rational residue and cancellation checks at six parameter triples through $N = 10$; and both symmetric finite-product coefficient laws through $N = 12$, including three coefficients beyond the expected truncation at each level.

Additional checks cover the printed A_1 and initial-head data, polynomial-weighted primitive telescoping, the factor- $1/2$ residue, an independent truncated-exponential coefficient calculation, and the product symmetrization in the Euler-primitive formula. All use exact rational or symbolic polynomial arithmetic. Their scope is finite regression verification; the all-order assertions are proved in the preceding sections, and are not proof-assistant formalizations.

11.3 Numerical diagnostics with independently constructed sides

The numerical suites completed **293 comparisons at 70 working decimal digits**. The central spectral tests use five parameter triples, $N = 0, 1, 2, 3$, and Taylor orders $0, 1, 2, 3$. Two series-side

evaluations use 80 actual terms with coefficients through A_{18} and 120 actual terms with coefficients through A_{22} . Both are compared to the finite Gamma–Stieltjes formula. The parameter triples include one and two simultaneous reciprocal-Gamma zeros.

For a head length M , the series-side approximation is

$$\sum_{n=0}^{M-1} \left[W_n(u) - \sum_{j < K} A_j(u) x_n^{-1-2u-2j} \right] + \sum_{j=K}^J A_j(u) \zeta(1+2u+2j, M+\kappa). \quad (11.1)$$

It uses the actual Gamma coefficient sequence, not the exact sum $G(u)$. Its Taylor coefficients are computed by ordinary truncated-series arithmetic. The three explicit harmonic series are also checked by a separate digamma/Bernoulli tail construction, rather than merely calling the same Dougall evaluator twice.

Other tests compare interior generating functions with independent hypergeometric/Lerch evaluations; check rational root-of-unity filters with order derivatives; differentiate the finite κ -primitives; and evaluate the continuous-shift and half-parameter primitive endpoints. The largest recorded scaled discrepancy was

$$2.18454477481895 \times 10^{-48}, \quad \text{scaled discrepancy} = \frac{|\text{left} - \text{right}|}{\max(1, |\text{right}|)}. \quad (11.2)$$

For the 120-term, A_{22} central spectral run it was $1.54137402015147 \times 10^{-58}$. These are observations about the executed computations, not certified counts of correct digits. Full values, parameter choices, tolerances and versions are in `results/numerical_checks.json`.

11.4 Reproduction and delivery scope

The tested environment used Python 3.13.5, SymPy 1.14.0 and mpmath 1.3.0. From the extracted package directory, run

```
python -m pip install -r requirements.txt
python code/verify_exact.py
python code/verify_numeric.py --part 1
python code/verify_numeric.py --part 2
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The numerical parts are separately executable to fit constrained tool runtimes; together they perform the same assertions as the default full run. The scripts regenerate their JSON reports. Reproduction changes runtime metadata, so it changes the corresponding file hashes. No network access is needed after installing the dependencies and TeX packages.

The numerical implementation supports the real parameter domain tested here and uses arbitrary precision controlled by mpmath. The article’s analytic continuation in u is broader than the set of numerical samples. The source audit, integration notes, manuscript-ready core section, result files, and SHA-256 manifest accompany the TeX and compiled PDF.

12 Further research questions

1. Classify cancellation operators beyond this family. The operator $E = \partial_u - \partial_\kappa - \partial_b - \partial_c$ acts on just one centered Gamma pair and produces a Hurwitz Bernoulli sequence. Determine

necessary and sufficient conditions on a product of complementary Gamma pairs for an analogous parameter vector field to collapse all resonance constant terms. An exact classification of this mechanism would be stronger than cataloguing isolated summations.

2. Compute multivariate resonant jets without path artifacts. The germ Q_N handles every reciprocal-Gamma zero along one spectral coordinate. Allow b and c to approach integral values simultaneously with $u + N \rightarrow 0$, and formulate a multivariate coefficient identity before specializing any coordinate. The normally convergent subtracted series supplies a natural holomorphic target. Determine which alternative finite-part prescriptions agree with that target and which introduce path-dependent constants.

3. Reduce higher harmonic-jet primitive endpoints. Equation (9.13) evaluates all primitive orders for the first symmetric spectral jet. For the second and higher jets, determine a closed generating system for the products of harmonic numbers and odd zeta tails that arise. A useful first target is a finite formula for every primitive of (7.8), with explicit control of whether double zeta values or only products remain. Such a reduction is not an independence statement about the resulting constants.

4. Integrate higher spectral jets in the center parameter. Theorem 10.1 solves the zeroth-order problem at every N . Its higher-jet counterparts contain Stieltjes functions and polygamma products. Develop normalized primitives through $(N, m) = (3, 2)$, then seek a finite all-order recursion. Keep additive constants and parameter-derivative conventions explicit.

5. Compare local endpoint and centered counterterm bases. At rational κ , the present counterterms are finite combinations of polylogarithms and their order derivatives. Construct their precise connection with local expansions in powers and logarithms of $1 - z$, including constant terms at both integral and half-integral spectral values. This would sharpen the comparison with classical hypergeometric continuation and partial-sum formulas.

6. Classify exact truncation loci. The symmetric products (6.4)–(6.5) and Proposition 6.2 are explicit loci where the centered asymptotic expansion terminates. Determine all parameter loci for which $W_n(u)$ is a Laurent polynomial in $n + \kappa$. Then classify which transverse jets retain finite reductions and which produce nontrivial zeta derivative towers.

7. Extract rational-level identities without nonreduction claims. Specialize the residue and jet formulas at small rational centers and parameters, and convert the Hurwitz values into compatible cyclotomic or Dirichlet-character coordinates. The goal is a short explicit identity catalogue, with each coefficient obtained exactly. Failure of a smaller numerical basis to reproduce a value is not a proof that the omitted coordinates are necessary.

8. Formalize the algebraic and analytic layers separately. The formal exponential recursion and differential law are polynomial identities suitable for a proof assistant. Their analytic interface is the compact-normal convergence of the subtracted Gamma-ratio series. Formalizing both would establish a machine-checked chain from classical Dougall summation to the complete jet formulas. The present package provides regression data, not that formalization.

13 Conclusion

The explicit Dougall target proposed in the incoming research programme has a complete answer in the centered ${}_5F_4$ family studied here. Complementary Gamma pairs enforce even inverse-power corrections; their Bernoulli coefficients obey a differential law; that law removes all lower counterterms from every integral-resonance constant. The residue polynomial and full Taylor formula then organize Stieltjes constants, negative-odd Hurwitz derivatives, and reciprocal-Gamma degeneracies in one ordinary convergent calculus.

The interlaced half-integer points explain a distinct nonpositive-even zeta tower. Harmonic specialization produces explicit identities with γ_1 , γ_2 and $\zeta'(-1)$, while beta differentiation evaluates an entire family of Euler-primitive endpoints. Polynomial weighting and the normalized Hurwitz ladder give antiderivatives at every resonance in the center parameter. These are exact structural results; the computational files make their finite coefficient and implementation checks reproducible without changing their proof status.

A Implementation details and coefficient conventions

The implementation stores ordinary Taylor coefficients. For arrays $a = (a_0, \dots, a_m)$ and $b = (b_0, \dots, b_m)$, multiplication is truncated convolution. If $\exp(L(t)) = \sum e_j t^j$, then

$$e_0 = e^{L_0}, \quad e_j = \frac{1}{j} \sum_{r=1}^j r L_r e_{j-r}. \quad (\text{A.1})$$

This recursion is used both for Gamma germs and for the coefficient generating function, with their different indices kept separate.

For $r_4 = r_4(u_0)$ and $\ell \geq 1$, the spectral Taylor coefficient of the logarithmic asymptotic coefficient is especially simple:

$$[t^\ell] \ell_j(u_0 + t) = -\frac{(-1)^\ell}{j(2j+1)} \binom{2j+1}{\ell} B_{2j+1-\ell}(r_4), \quad 1 \leq \ell \leq 2j+1. \quad (\text{A.2})$$

It vanishes beyond that degree. Thus all A_j jets can be generated by finite rational arithmetic at rational expansion points, without symbolic Gamma manipulation.

For the harmonic checks, the independent expansion

$$\psi(x + \tfrac{1}{2}) - \log x \sim -\sum_{j \geq 1} \frac{B_{2j}(\frac{1}{2})}{2j x^{2j}} \quad (\text{A.3})$$

is used at $x = n + 1/2$. Squaring it together with the logarithmic cross term gives the second harmonic tail. More generally,

$$\psi(n+1) - \log(n+a) \sim \sum_{k \geq 1} \frac{(-1)^{k+1} B_k(1-a)}{k(n+a)^k}. \quad (\text{A.4})$$

These are finite asymptotic approximations in the diagnostics, never asserted to converge as infinite series at fixed n .

The numerical code asks mpmath for spectral derivatives through its `zeta(..., derivative=r)` interface, rather than differentiating a special-value shortcut that may have a different numerical code path. The analytic formulas in the article remain the source of truth. The manifest covers the delivered files; neither a checksum nor a passing regression suite substitutes for review of the proofs.

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